CSST - Complication Signal Surveillance Template

Review identification, purpose and sources

v2.0

Study / protocol	CIP version	Site / code	CSST review ID / revision
Review type ☐ Scheduled ☐ Triggered	**Review period / data cut-off**	**Device / procedure reviewed**	**Previous CSST review ID**
SCML snapshot ID	**Analysis file (if used)**	**Prepared by / role**	**Controlled location**

A Purpose and use

The CSST is the site's review of its complications taken together. A single event is assessed in the SART and indexed in the SCML; the CSST looks at all of them at once, on the site's own schedule and after a trigger, to see whether events that each seemed explicable share a cause: the same component, the same surgeon, the same moment of follow-up or the same kind of case. A concentration of events is a question to test, not a conclusion about a device or a surgeon.

Unit of analysis	Decision supported	Inputs	Outputs
One review of the site's complications for one period, or one review opened by a trigger.	Whether the data show no material pattern, a pattern to watch more closely, a signal to send on, or a result limited by the data.	SCML events, ICDL procedures and context, PFVR follow-up reached, SLCT surgeon sequence, SART event status and the previous CSST.	Counts by time point (C) Triggers met (D) Signals assessed (F) Routes to the receiving records (H)

B Sources and their limitations

With limitation means the review goes ahead: name the records that are missing or incomplete. Not available means the counts that need that source cannot be made in this review.

Source	Used for	ID · extraction date	Status	Limitation (records missing or incomplete)
SCML snapshot	Events and their categories		☐ Complete ☐ With limitation ☐ Not available	
ICDL procedure records	Procedures, components, configuration, context		☐ Complete ☐ With limitation ☐ Not available	
PFVR / visit schedule	Participants who reached each follow-up time point		☐ Complete ☐ With limitation ☐ Not available	
SLCT	Surgeon codes and case sequence		☐ Complete ☐ With limitation ☐ Not available	
SART	Event type, seriousness and status		☐ Complete ☐ With limitation ☐ Not available	
Previous CSST review	Signals to follow up in G		☐ Complete ☐ With limitation ☐ Not available	

C1 Cases and follow-up reached

These numbers are the denominators for C2 and C3. Procedures and surgeons come from the ICDL and the SLCT, enrollment from the PSEAR, follow-up from the PFVR. The time points match the columns of C2. Adapt the time points to the CIP visit schedule.

Count	This period (n)	Previous period (n)	Change (%)	Cumulative (N)
Participants enrolled				
Investigational procedures performed				
Surgeons operating				
Surgeons with a first study case in this period				
Procedures within a surgeon's first ___ study cases				
Revision procedures (component exchanged or removed)				
Explanted components returned				
Follow-up visits due				
Follow-up visits done in window				
Follow-up visits missed				
Participants who reached each time point (N for the columns of C2)				
Operated (intraoperative)				
Reached discharge				
Reached first postoperative visit				
Reached 6 weeks				
Reached 3 months				
Reached 6 months				
Reached 12 months				
Reached 24 months				
Participants withdrawn or lost to follow-up				

CSST - Complication Signal Surveillance Template

Complications by time point · 1 of 2

v2.0

C2 Complications by follow-up time point

Count each event once, in the time point where it occurred, against the N of that column. Reading across a row shows whether a problem appears early or grows with time.

Event (SCML category)	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	Grade IIIa or higher (n)	New this period
N for this column (from C1)											
Instability / dislocation											
Dislocation											
Subluxation or instability symptoms											
Fracture											
Intraoperative fracture											
Periprosthetic fracture											
Acromion or scapular spine fracture											
Infection / wound											
Superficial wound problem											
Deep or periprosthetic infection											
Hematoma											
Neurologic / vascular											
Nerve injury (axillary nerve, brachial plexus)											
Vascular injury											
Device performance											
Component loosening											
Component disassociation (liner, glenosphere)											
Implant breakage or wear											
Radiographic change (notching, lucency, migration)											

C2 Complications by follow-up time point (continued)

Event (SCML category)	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	Grade IIIa or higher (n)	New this period
N for this column (from C1)											
Function and other											
Stiffness or loss of motion											
Persistent pain											
Thromboembolic event											
Other (SCML code): _____											

Overview

Measure	This period (n)	Previous period (n)	Change (%)	Cumulative (n / N) %	Cumulative at previous review (n / N) %
Participants with at least one complication					
Participants with a complication of grade IIIa or higher					
Participants with more than one complication					
Reoperations					
Revisions (component exchanged or removed)					
Serious events (SAE / SADE)					
Device deficiencies (all; counted apart from complications)					

CSST - Complication Signal Surveillance Template

Complications by grade

v2.0

C2 Complications by Clavien-Dindo grade and time point

Grades come from the SCML, which copies them from the SART. Count each complication once, in the column of its time point. Device deficiencies without an adverse event are not graded.

Grade	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	New this period
N for this column (from C1)										
I · No treatment beyond simple care										
II · Drug treatment, transfusion										
IIIa · Intervention without general anesthesia										
IIIb · Intervention under general anesthesia										
IVa · Life-threatening, single organ										
IVb · Life-threatening, multiorgan										
V · Death										
Disability at discharge (suffix d)										

C3 Procedure context

For each marker, compare the complication rate in procedures with it and without it. Context never changes the counts in C2; it helps explain them.

Context marker	Procedures with it (n / N) %	With it: had a complication (n / n) %	Without it: had a complication (n / n) %	Source
Unplanned configuration change				
Component substitution				
Technical deviation from the surgical technique				
Intraoperative difficulty (bone quality, deformity, revision case)				
Senior support or proctoring used				
Surgeon within first ___ study cases				
Operation outside normal hours				
Instrument or loaner set problem				
Gap in the operative record				
Other: _____				

Links between context and complications

Context marker	Procedures (ICDL IDs)	Complication found (C2 event · SCML ID)	Comment

E Comparison by surgeon, component and case type

Grades follow Clavien-Dindo (SCML). A group stands out when its share of grade IIIa–V complications is clearly larger than its share of procedures. Use the surgeon codes of the SLCT and the component, lot and configuration recorded in the ICDL.

E1 · By surgeon (SLCT code)

Surgeon	Procedures (n) % of all	Grade I–II (n / n) %	Grade IIIa–V (n / n) %	Share of all IIIa–V %	IIIa–V in first ___ cases (n / n) %	Note

E2 · By component, lot or configuration (ICDL)

Component · lot · configuration	Procedures (n) % of all	Grade I–II (n / n) %	Grade IIIa–V (n / n) %	Share of all IIIa–V %	Note

E3 · By case type

Case type (indication, primary or revision)	Procedures (n) % of all	Grade I–II (n / n) %	Grade IIIa–V (n / n) %	Share of all IIIa–V %	Note

D Signal triggers

Apply each trigger and write the value found. A trigger met opens a signal in F.

ID	Trigger	What is compared · where	Agreed limit	Source of the limit	Value found (n / N) %	Met?
T1	**Rate above the agreed limit**	Any category at any time point; one limit for all grades and a lower one for grade IIIa–V (C2).				☐ Yes ☐ No ☐ Not assessable
T2	**Rate rising with time**	The same category grows from earlier to later time points (C2 row).				☐ Yes ☐ No ☐ Not assessable
T3	**Rise since the previous review**	Same category and time point, compared with the previous CSST.				☐ Yes ☐ No ☐ Not assessable
T4	**New category**	A category with no earlier events now has events (C2, New this period).				☐ Yes ☐ No ☐ Not assessable
T5	**Surgeon share**	A surgeon's share of grade IIIa–V complications exceeds their share of procedures (E1).				☐ Yes ☐ No ☐ Not assessable
T6	**Component or configuration share**	Same comparison for grade IIIa–V by component, lot or configuration (E2).				☐ Yes ☐ No ☐ Not assessable
T7	**Early learning cases**	Grade IIIa–V rate in a surgeon's first cases above their later cases (E1, C1).				☐ Yes ☐ No ☐ Not assessable
T8	**Context association**	Rate with a context marker above the rate without it (C3).				☐ Yes ☐ No ☐ Not assessable
T9	**Serious events grouped**	Serious events in one category within a short window (SART, C2).				☐ Yes ☐ No ☐ Not assessable
T10	**Device deficiencies of one type**	Repeated deficiencies of the same type or component (C2, SART).				☐ Yes ☐ No ☐ Not assessable
T11	**Same severe complication repeated**	The same complication of grade IIIa–V in two or more participants (C2).				☐ Yes ☐ No ☐ Not assessable

These triggers are examples. Before the first inclusion, review them and set each limit with the site's clinical team and, where it applies, the sponsor; write the agreed limit and its source (CIP, sponsor guidance, site procedure). A limit changed later applies only to later reviews.

F Assessment of each signal

One row per trigger met. Describe the pattern and the explanations still to test; do not conclude the cause. Set how the next review will show whether the response worked.

Signal ID	Trigger (D) · events (SCML IDs)	Pattern described	Explanations to test	Disposition	What will show the response worked
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	

CSST - Complication Signal Surveillance Template

Earlier signals and routes

v2.0

G Signals from earlier reviews

Check each earlier signal against the criterion set for it in F.

Signal ID · review	Criterion set then	Result in this review	Outcome	Next step
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	

H Where each signal goes

The action is followed in the receiving record; this review records only that it was sent.

Route	Question for this review	Signal ID (F)	Receiving record · date sent	Status
SLCT	Could surgeon-dependent variation explain the pattern?			☐ Sent ☐ Not required ☐ Pending
Sponsor device-deficiency route	Does the signal point to a device, component, configuration or lot?			☐ Sent ☐ Not required ☐ Pending
SART	Are the individual events complete and routed through the safety pathway?			☐ Sent ☐ Not required ☐ Pending
SRR	Does the signal show a local process weakness to enter as a risk?			☐ Sent ☐ Not required ☐ Pending
CAPA log	Did a first response fail, or does a process failure recur?			☐ Sent ☐ Not required ☐ Pending
POT	Does the PI need to restrict, supervise more, continue or escalate?			☐ Sent ☐ Not required ☐ Pending
PIR / IMC	Should the result inform an indicator or an internal check?			☐ Sent ☐ Not required ☐ Pending

| Conclusion and sign-off

The conclusion reflects the most serious open signal.

Review conclusion	☐ No material pattern ☐ Closer surveillance ☐ Signal sent on ☐ Inconclusive: data limitation
Next review · date and what will be checked again	

Limitations and rationale

Reviewer

Name / role:

Signature / date:

PI / accountable approver

Name / role:

Signature / date: